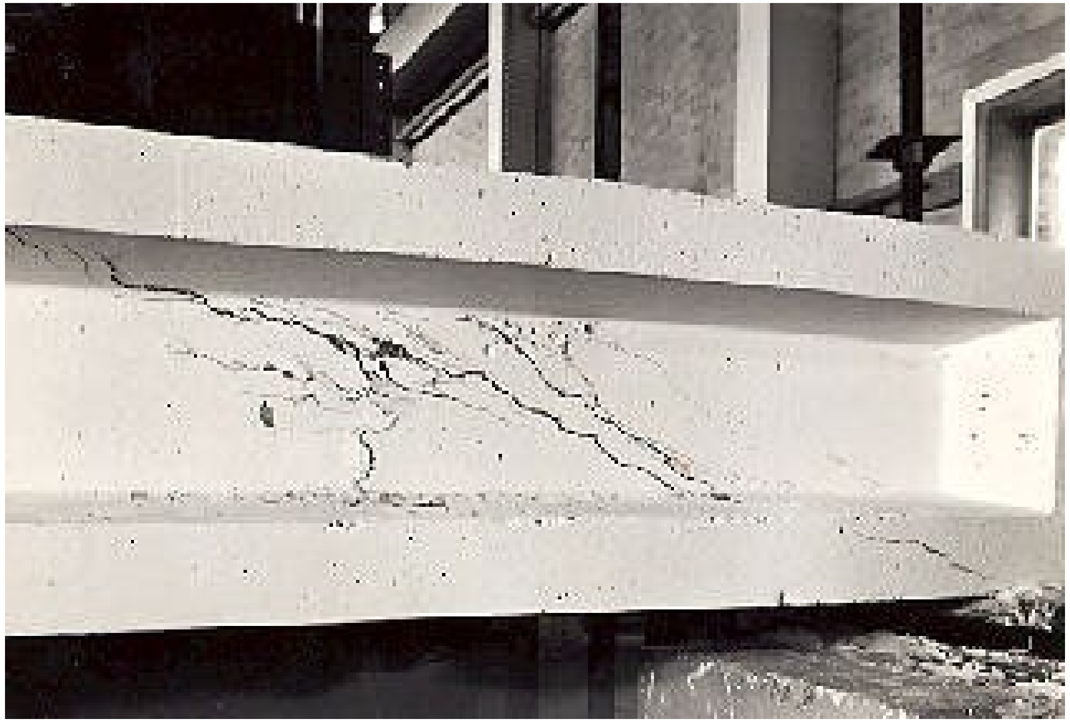
Analysis and Design of simple PC Beams - Ultimate limit State

Reference Text – Prestressed Concrete

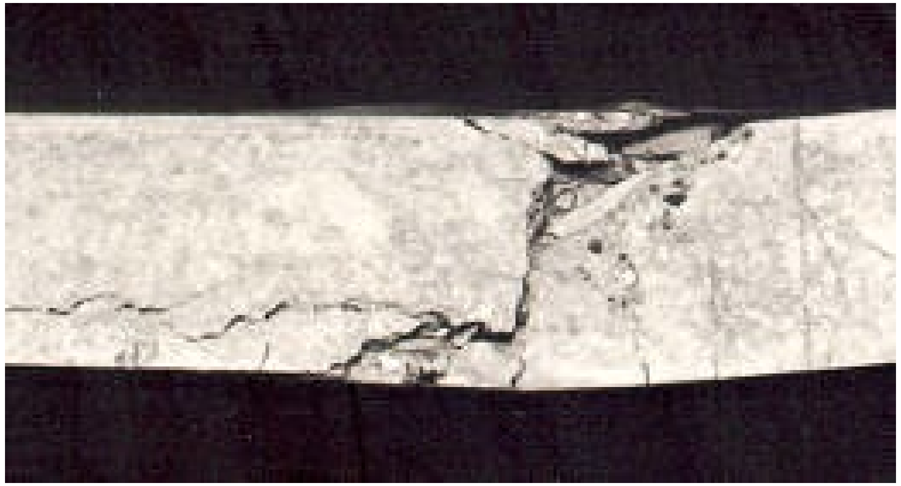
Chapter 7 Shear

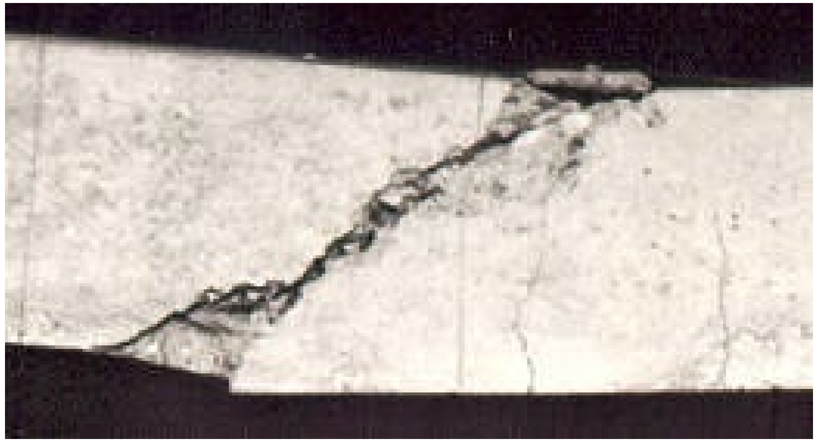
Ultimate limit State

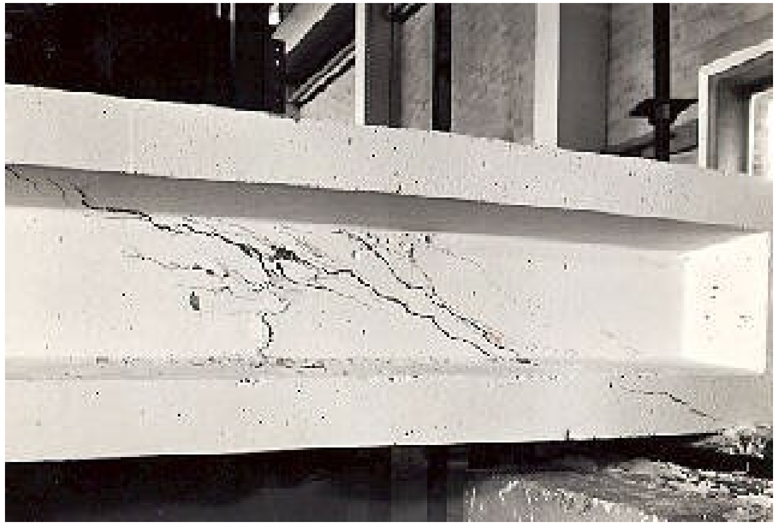
SHEAR Strength of Prestressed Concrete Beams



Revision Shear in RC Beams

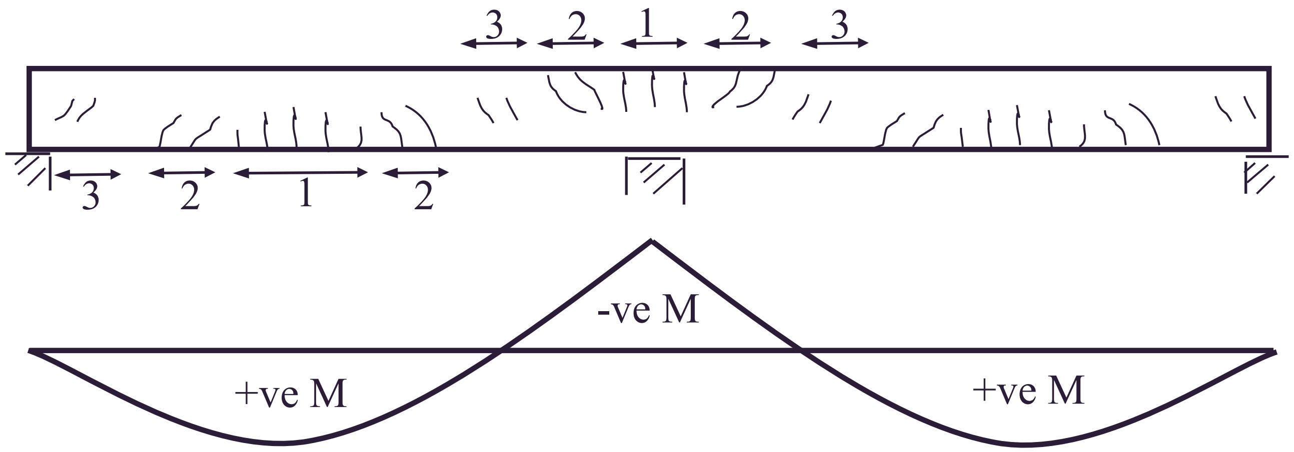




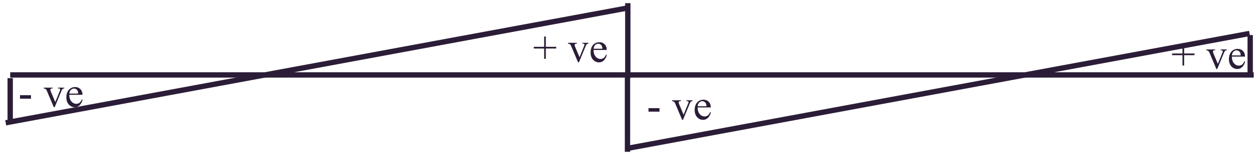


Cracking pattern of R/C Beams at Failure

Consider the two span continuous beams



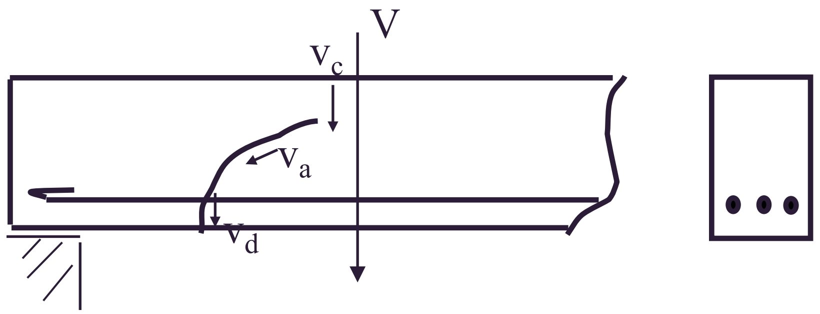
Bending moment diagram



Shear force diagram

- Zone 1 – Flexural Cracks only, bending moments large, shear small
- Zone 2 – Flexural-Shear cracks, bending moments large, shear force significant inclined cracks
- Zone 3 – Web-Shear cracks (**diagonal tension cracks**), shear force large, bending moment small

Load carrying mechanisms
Shear transfer in concrete at inclined crack



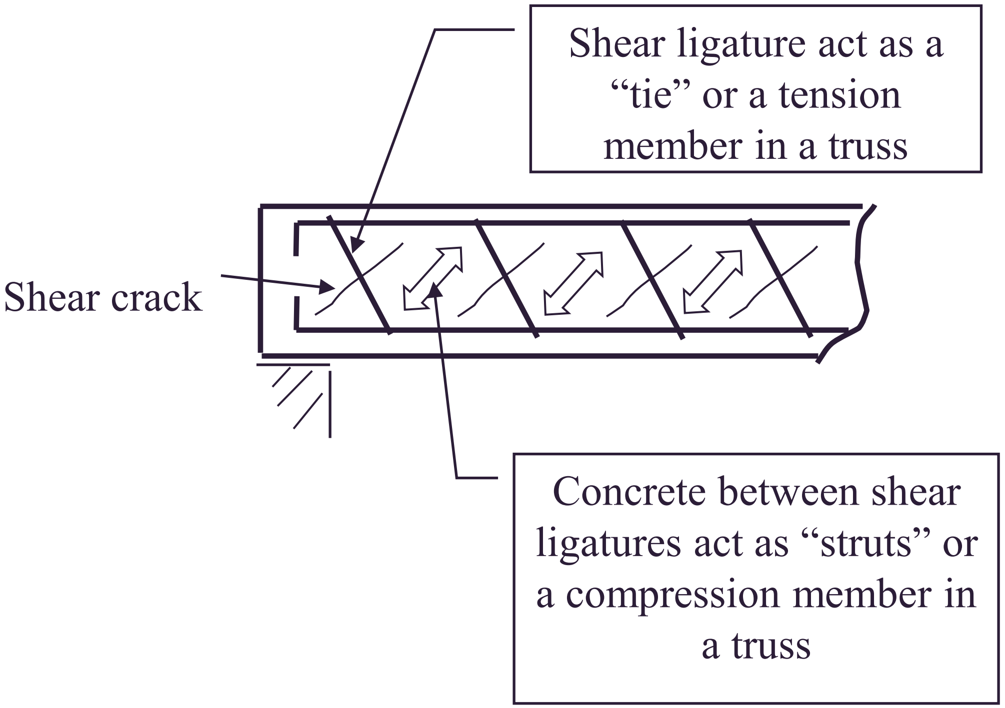
Along the section of the crack the shear force V is resisted by a combination of the following

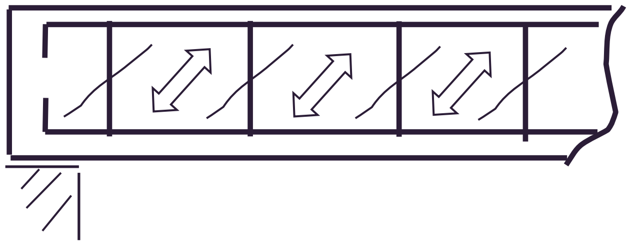
Concrete shear stress v_c over the uncracked zone (ie compressive concrete above the tip of the crack)

Vertical component of the frictional interlock v_a between the two jagged faces each side of the crack

A dowel force V_d provided by the longitudinal tensile reinforcement

Web Reinforcement contribution to Shear Capacity
Truss analogy





Inclined Shear Ligatures

Structurally most efficient
Seldom used because hard to place

Vertical Shear Ligatures

Common in practice
Easy to analys, easy to place

Simple Truss analogy

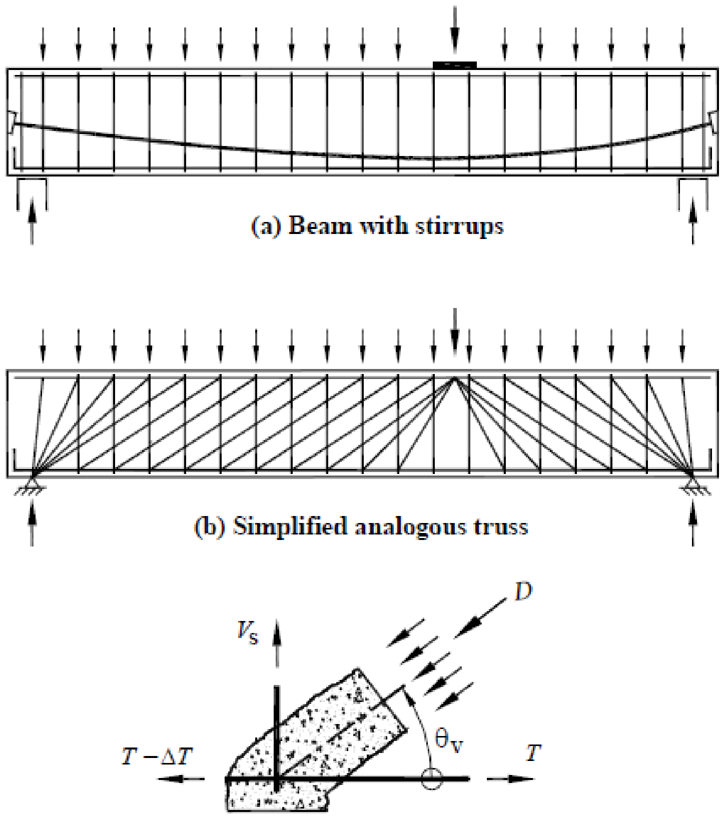
$$V_s = D\sin\theta_v + P_v$$

$$\Delta T = D\cos\theta_v$$

V_s tensile force carried by a stirrup

D is the diagonal force in the inclined compressive concrete

T is the total tensile force in the stringer

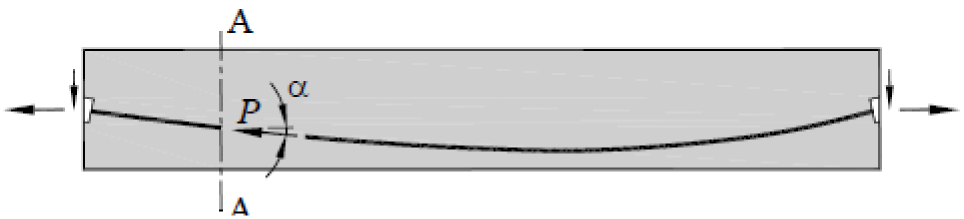


(c) Joint forces: lower joint in analogous truss

Effect of prestress on behaviour in Shear

Prestressing has a significant effect on the shear at diagonal cracking and on the load capacity in shear.

The curved profile of prestressed cables in concrete members produces horizontal and vertical component of prestress on the section



Effect of horizontal prestress component

Delays the formation of flexural and inclined cracks.

Web-shear cracks occur more frequently in PT members since:

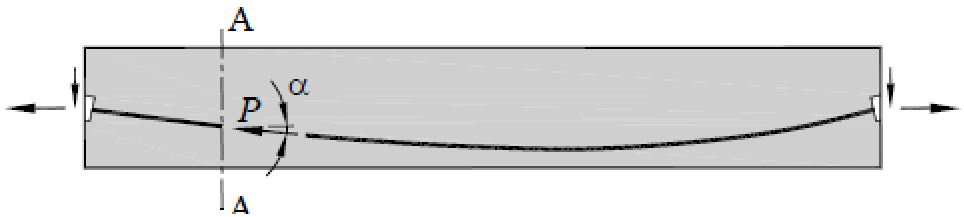
- the sections have thinner webs therefore higher shear stresses than RC members,
- horizontal prestress inhibits flexural cracking in bottom fibres.

Shear Force at web-shear cracking (diagonal cracking)

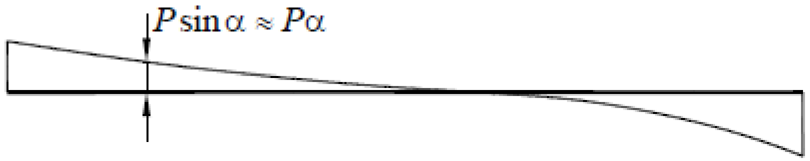
Effect of vertical prestress component

Magnitude of shear force acting on the concrete affected by vertical component $P\sin\alpha$ (α = cable slope)

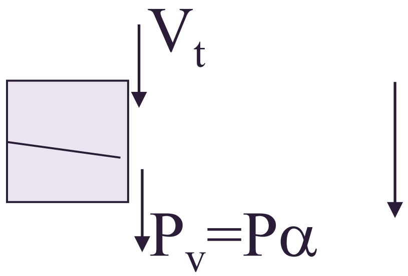
Beam with parabolic cable



Shear force due to cable slope



Effect on web-shear cracking

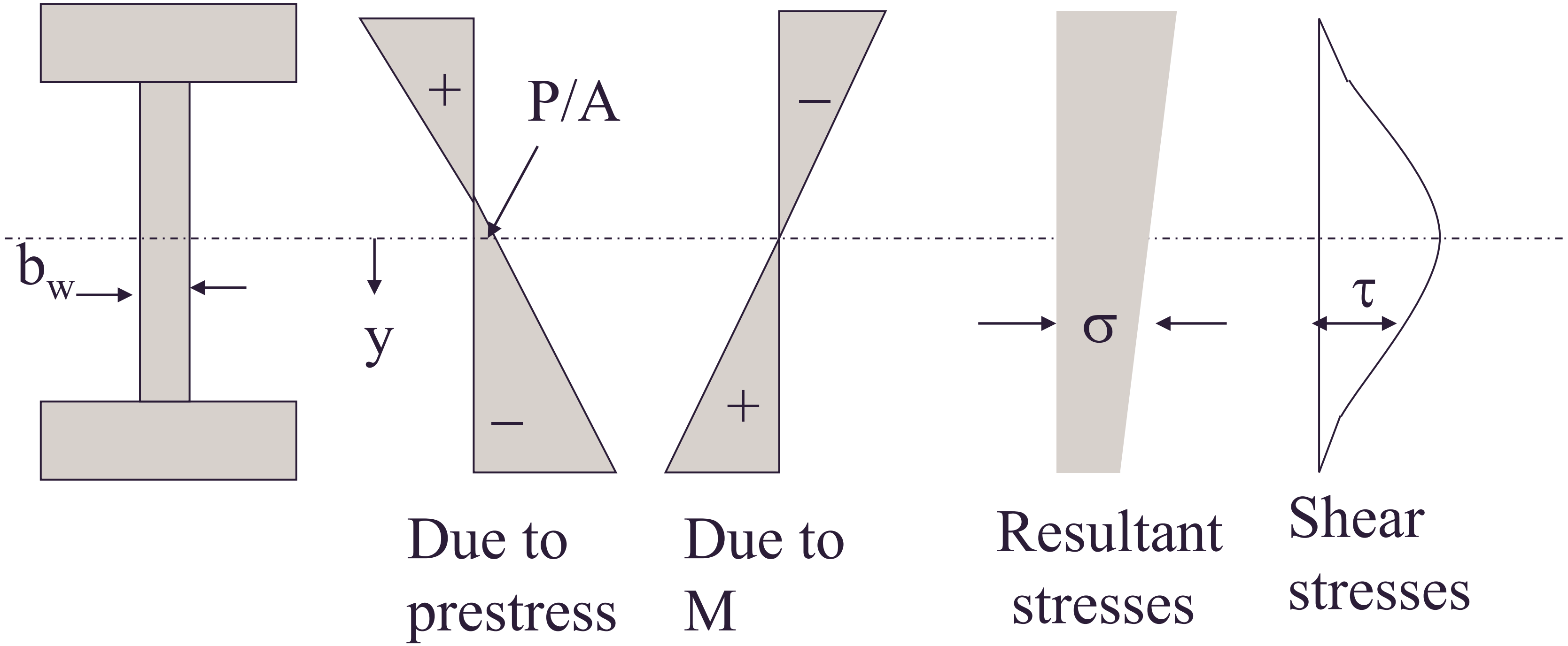


Shear force in the concrete due to web-shear cracking:

$$V_{cr} = V_t + P_v$$

Web-Shear cracking

Web-shear cracking load can be estimated by equating principal tensile stress σ at a critical point in the web to the tensile strength of concrete.



$$\sigma = -\frac{P}{A} - \frac{Pey}{I} + \frac{My}{I}$$

$$\tau = \frac{V_t Q}{I b_w}$$

Q is the 1st moment of area above (or below) centroid

Equation for Principal stresses

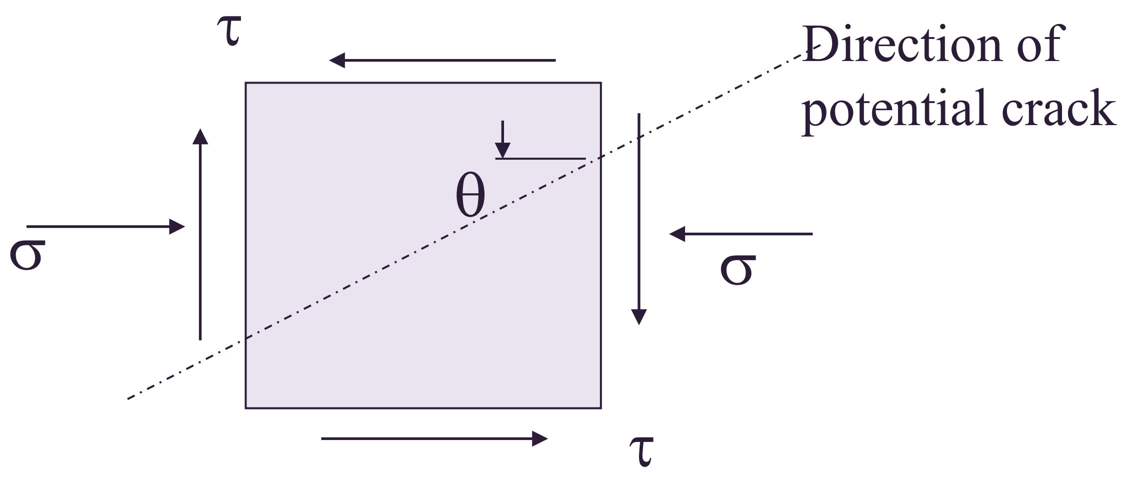
$$\sigma_{\max/\min} = \left(\frac{\sigma_x + \sigma_y}{2}\right) \pm \sqrt{\left(\frac{\sigma_x - \sigma_y}{2}\right)^2 + \tau_{xy}{}^2}$$

In this problem it is assumed that $\sigma_y = 0$

σ is taken as positive if tensile or negative if compressive

Web-Shear cracking

Stresses acting on element



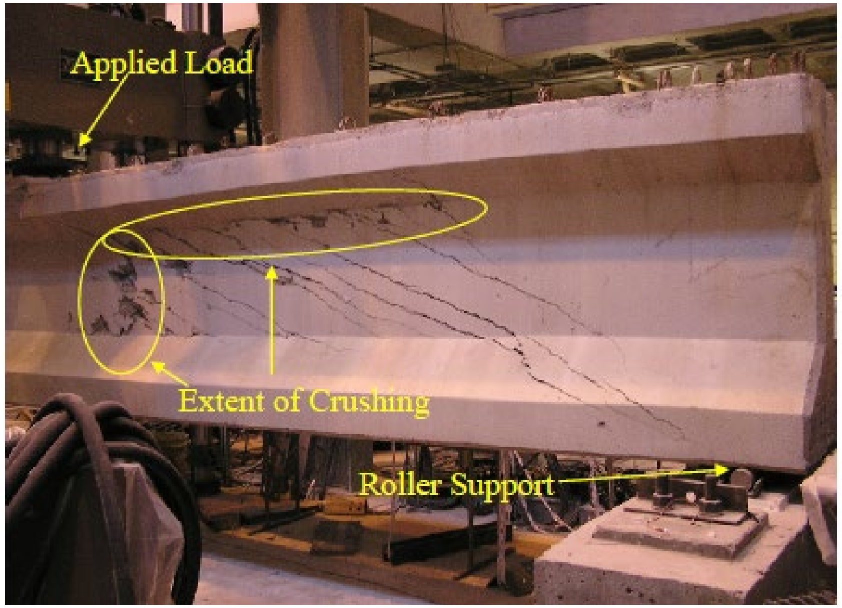
- Principal tensile stress σ_1 is given by

$$\sigma_1 = \sqrt{(0.5\sigma)^2 + \tau^2} + 0.5\sigma$$

- In AS3600 a principal tensile stress of $0.36\sqrt{f'_c}$ is assumed sufficient to cause diagonal cracking
- The corresponding shear force V_t is found by setting σ_1 equal to this value and substituting in equations for σ and τ

Shear force in the concrete due to web-shear cracking:

$$V_{cr} = V_t + P_v$$



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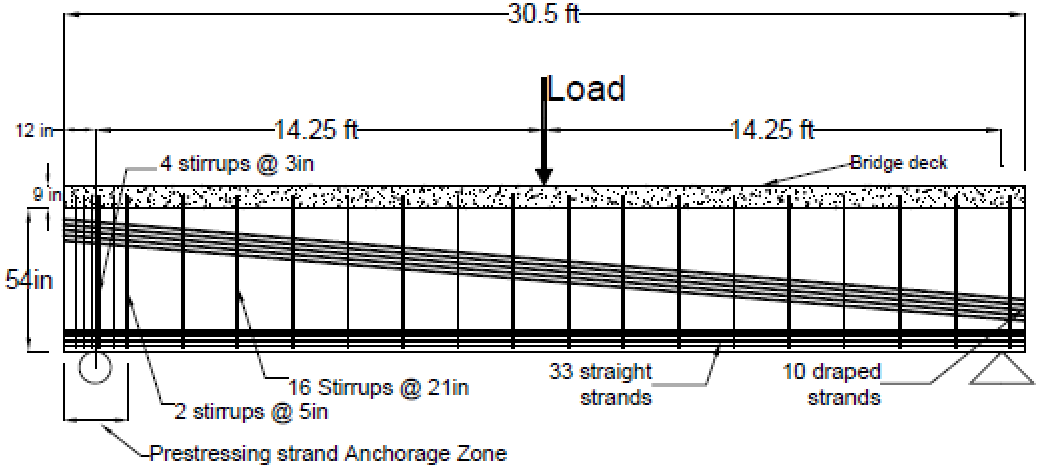
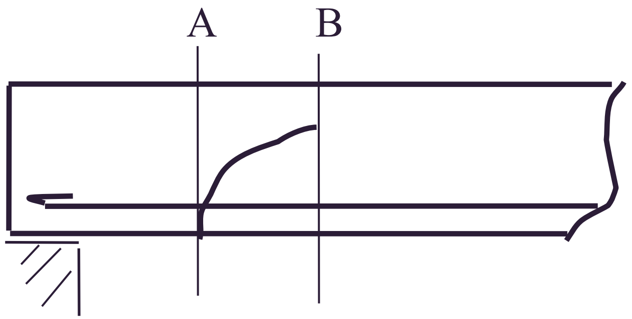
TYPICAL SHEAR FAILURE IN PSC GIRDERS



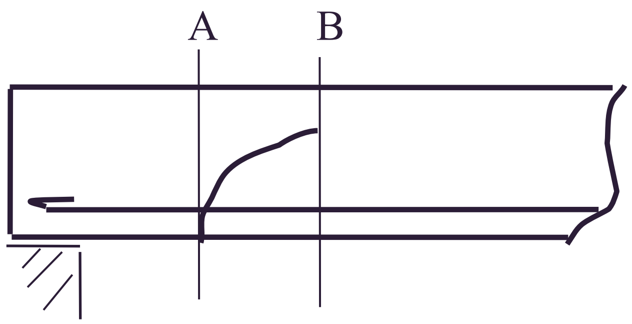
Total Shear Capacity, V_u

$$V_u = V_{uc} + V_{us} + \{P_v\}$$



- V_{uc} represents combined effects of aggregate interlock, dowel action and shear in compressive concrete above the tip of the crack (lumped together)
- V_{us} is truss action term for the effect of shear stirrups
- P_v vertical component of prestress (either increasing or decreasing the section capacity or as a load on the member)

Contribution of Concrete to Shear Strength, V_{uc}



$$V_{uc} = k_v b_v d_v \sqrt{f'_c}$$

Cl8.2.4.1 of AS3600

- k_v parameter that determines capacity of the web to resist stresses that provide concrete contribution to shear strength, k_v and concrete strut angle, θ_v determined from the general method given in Cl8.2.4.2.
- d_v internal moment lever arm at critical cross section
 (greater of 0.72D or 0.9d is the distance from the extreme compression fibre to the centroid of the longitudinal tension reinforcement)
- b_v effective width of the web.

In determining bv at a particular level, for prestressing ducts with $d_d > b_w/8$:

$b_v = (b_w - k_d \Sigma d_d)$ . . where

Σd_d = sum of the diameters of prestressing ducts, if any, in a horizontal plane across the web

k_d = 0.5 for grouted steel duct;0.8 for grouted plastic duct; 1.2 for ungrouted duct

Contribution of Concrete to Shear Strength, V_{uc}

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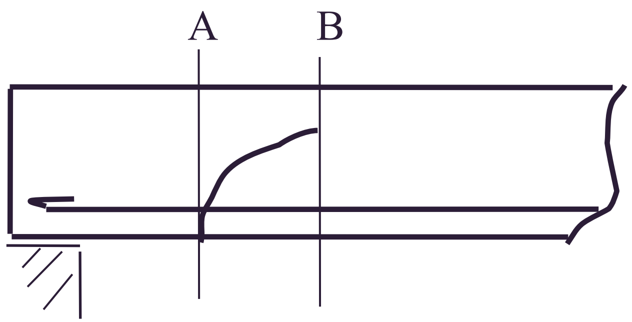
$$V_{uc} = k_v b_v d_v \sqrt{f'_c}$$



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  k_d = 0.5 for grouted steel duct;0.8 for grouted plastic duct; 1.2 for ungrouted duct

General method for k_v & θ_v , Cl8.2.4.1

For k_v

$\frac{A_{sv}}{s} < \frac{A_{svmin}}{s}$, then $$k_v = \left[\frac{0.4}{1+1500\varepsilon_x}\right]\left[\frac{1300}{1000+kdgdv}\right];$$

$\frac{A_{sv}}{s} \geq \frac{A_{svmin}}{s}$ $$k_v = \left[\frac{0.4}{1+1500\varepsilon_x}\right]$$

Where **$f`_c \leq$ 65MPa** $k_{dg} = \left[\frac{32}{16+dg}\right]$; *but not less than 0.8 ;* d_g *is maximum aggregate size*

for $f`_c \geq$ 65MPa $k_{dg} = 2.0$

Estimate θ_v

$\theta_v = 29 + 7000\varepsilon_x$

Estimating ε_x (longitudinal strain in concrete for shear at mid-depth of the section)

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$$\varepsilon_x = \frac{|M^*/d_v| + |V^*| - gP_v + 0.5N^* - A_{pt}f_{po}}{2(E_sA_{st} + E_pA_{pt})}; \textit{within the limits } 0 \le \varepsilon_x \le 3x10^{-3}$$

where ε_x above is less than zero ex can be taken as zero of calculated as follows

$$\varepsilon_x = \frac{|M^*/d_v| + |V^*| - gP_v + 0.5N^* - A_{pt}f_{po}}{2(E_sA_{st} + E_pA_{pt} + E_c\,A_{ct})}; \textit{within the limits } -0.2x10^{-3} \le \varepsilon_x \le 0$$

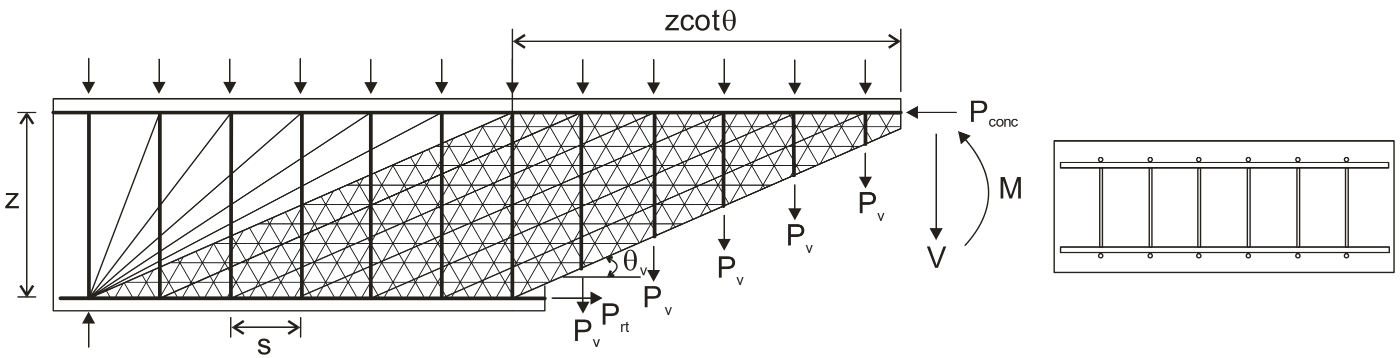
where **M*** and **V*** are absolute design action bending moment and shear force values and ***M** $\geq$ *(V*-P$_v$)d$_v$***; note: M* here is at the same location of V* and it need **NOT** be the maximum bending moment of the beam.

N* is positive(+) for tension and negative (-) for compression

A$_{st}$ & ***A$_{pt}$*** are the area of flexural tension reinforcing bars and prestressing tendons located in the half depth of section(between D/2 & extreme tensile fibre)

- f_{po} is the stress in prestressing tendons when strain in the surrounding concrete is zero.
- A_{ct} is the area of concrete between the mid depth of the section and the extreme tensile fibre.
- f_{po} may be taken as $0.5f_{pb}$ for bonded tendons outside the transfer length and σ_p for unbonded tendons.

Shear strength of beams with shear reinforcement - V_{us}



If all stirrups carry the same load (assumes uniform crack width)

$$V = nP_v$$

And failure occurs at yield $V_{us} = nA_{sv}f_{sy.f}$ ← Force = number of stirrups x stirrup force

From geometry $n = z\cot\theta_v/s$ ← Force in stirrups = total area of stirrups x yield stress

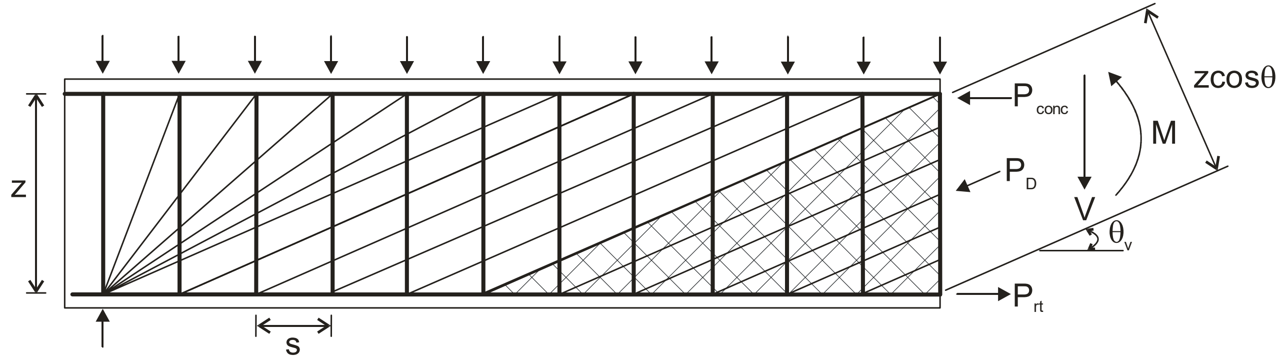
Hence $V_{us} = \frac{A_{sv}}{s}f_{sy.f}z\cot\theta_v$

Number of stirrups crossing a crack depends on the angle of the diagonal crack and the stirrup spacing

Note: $\boldsymbol{z = d_v}$ according to AS3600-2018

Cl8.2.5.2 $$\boldsymbol{V_{us} = \frac{A_{sv}}{s}f_{sy.f}\cot\theta_v d_v}$$

Shear strength due to web crushing



To check the possibility of failure of the concrete struts the stress in each strut must be determined

If the force in each strut is P_D so that the total force is nP_D where n=the number of stirrups exposed by the cut at θ_v

$$V_{us} = nP_D \sin\theta_v$$

Shear strength of an RC beam limited by concrete strut failure.

In order to apply this limit a failure criteria must be chosen

In AS 3600 Cl8.2.3.3 this is applied indirectly by placing an upper limit on the shear $V_{u.max}$ such that $V_u \leq V_{u.max}$ where

$$V_{u.max} = 0.55\left[0.9 f_c' b_v d_v \left(\frac{\cot(\theta_v) + \cot(\alpha_v)}{1 + \cot^2(\theta_v)}\right)\right]$$

This ensure failure is by yielding of stirrups and not by concrete crushing

Steps in the design of shear reinforcement – prestressed concrete

$$\phi V_u = \phi(V_{uc} + V_{us}) \geq V^* - \gamma_p P_v$$

The following steps can be used to determine the stirrup requirement at a given cross section:

1. Determine the design shear force V^* *and design* M^*

2. Calculate min allowed mid-height strain ε_x from Cl 8.2.4.2.2. Choose value of ε_x between calculated and maximum allowed +0.003 (starting values partially prestressed

3. With chosen ε_x calculate shear resisted by concrete V_{uc} from Cl 8.2.4.1

4. Calculate θ_v from Cl 8.2.4.2.1. angle θ_v taken as design variable between a minimum value determined with lowest possible ε_x, and 50°

5. Calculate $V_{u.max}$ from Cl8.2.3.3. if $V^*-\gamma_p P_v > \varphi V_{u.max}$ it is necessary to increase dimensions of the section

6. Recalculate the shear V_{uc} resisted by the concrete from Cl 8.2.4.1. If $V^* \leq \varphi V_{uc}$, then no further design is necessary except if D>750mm in which case $A_{sv,min}$ provided and calculated from Cl 8.21.7 $A_{sv.min} = 0.8 b_v s \sqrt{f'_c} / f_{sy.f}$

7. If $V^* > \varphi V_{uc}$ or if D>750mm, calculate A_{sv}

8. Calculate A_{sv}/s from Cl8.2.5.2 in which $V_{us} = \frac{1}{\varphi}[V^*-(\varphi V_{uc} + \gamma_p P_v)]$. Select stirrup size and spacing

EXAMPLE 7.1

CALCULATE THE LOAD ON A POST-TENSIONED BEAM IN SHEAR FOR A WEB-SHEAR CRACK TO FORM

A post-tensioned beam with the cross-section shown in Figure 7.8 is to span 25 metres between simple supports and carry a uniform live load of 24 kN/m. The beam is prestressed at each end by a parabolic cable with eccentricity of 524 mm at mid-span and zero at the ends. The cable comprises 22 strands of 12.7 mm diameter with $A_p = 2170\ \text{mm}^2$, in a 112 mm diameter duct. Owing to friction the prestressing force varies in an approximately linear manner from 2435 kN at each end to 2300 kN at mid-span.

The beam also has longitudinal reinforcement consisting of 5-N28 bars ($A_{st} = 3100\ \text{mm}^2$) at depth $d_{st} = 1160$ mm, throughout the span. For the longitudinal reinforcing steel and the stirrups we take $f_{sy} = 500$ MPa and $f_{sy.f} = 250$ MPa (Grade R). The concrete strength is $f_c' = 40$ MPa.

Calculate the shear corresponding to the formation of a significant web-shear crack at a distance of $x = 1.0$ metres from the support.

Other data:

$$A_g = 488.4\times10^3\ \text{mm}^2;\ I_g = 83.8\times10^9\ \text{mm}^4$$

$$y_t = 552\ \text{mm};\ y_b = 668\ \text{mm}$$

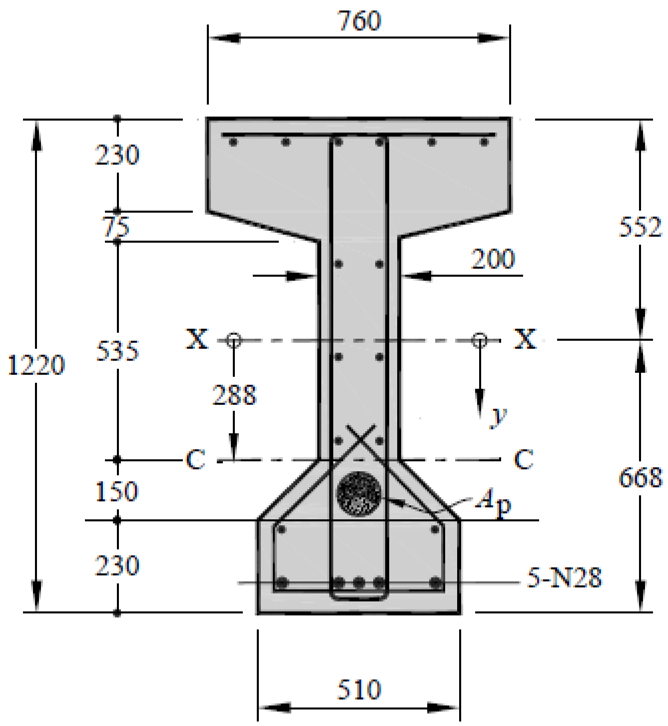


Figure 7.8 Cross-section for Examples 7.1, 7.2 and 7.3

EXAMPLE 7.2

ANALYSIS OF SHEAR STRENGTH AT A SECTION

The post-tensioned beam of Example 7.1, with the cross-section shown in Figure 7.8, is to carry a uniformly applied load along its full length. As before, the beam is prestressed at each end by a parabolic cable with eccentricity of 524 mm at mid-span and zero at the ends. The cable comprises 22 strands of 12.7 mm diameter with $A_p = 2170\ mm^2$, in a 112 mm diameter duct. Owing to friction the prestressing force varies in an approximately linear manner from 2435 kN at each end to 2300 kN at mid-span. The beam also has longitudinal reinforcement consisting of 5-N28 bars ($A_{st} = 3100\ mm^2$) at depth $d_{st} = 1160$ mm, throughout the span

If the member contains 2-leg R10 stirrups spaced at 270 mm centres throughout, calculate the shear strength of a section located at distance of $x = 1.0$ metres from the support.

Other data:

$f_c' = 40$ MPa; $f_{sy} = 500$ MPa; $f_{sy.f} = 250$ MPa; $f_{pb} = 1870$ MPa

$b_v = 144$ mm; $d_v = 878$ mm; $P_v = 187$ kN; $E_p = 195$ GPa

(note: b_v, d_v and P_v are calculated in Example 7.3).

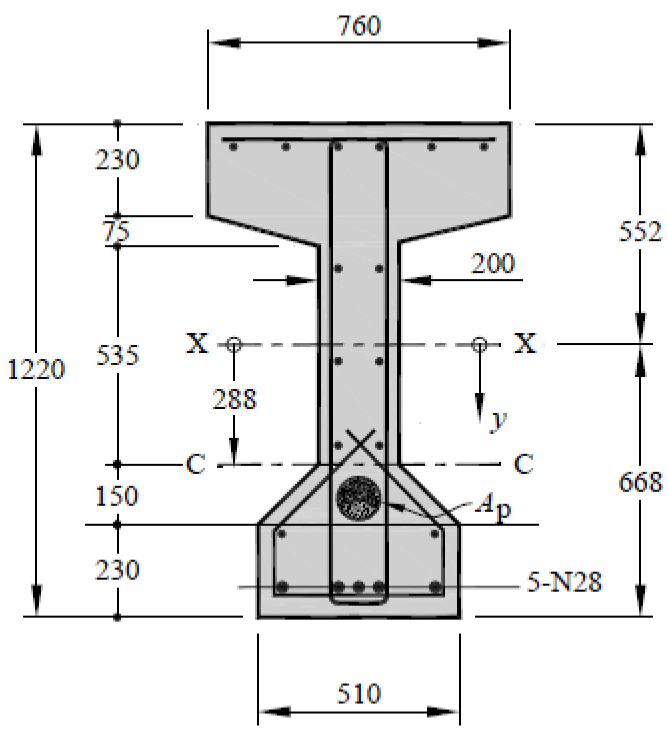


Figure 7.8 Cross-section for Examples 7.1, 7.2 and 7.3

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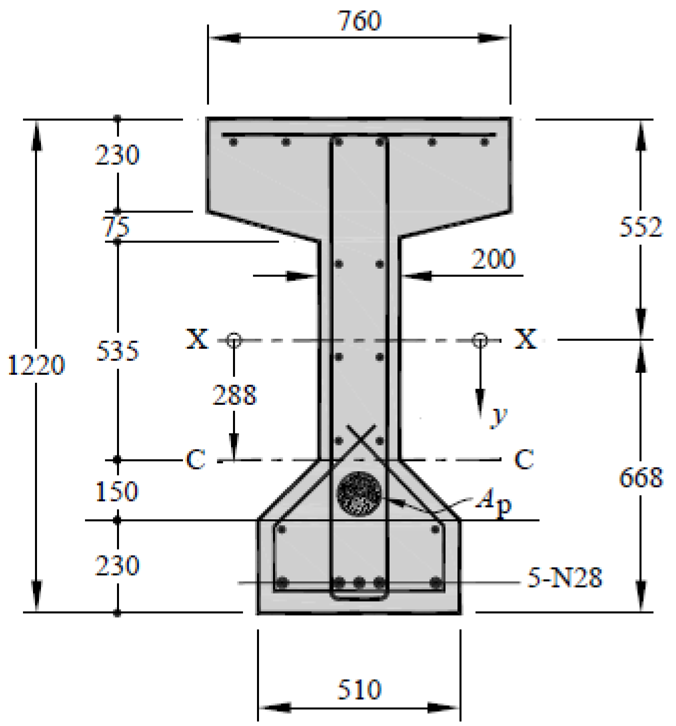
EXAMPLE 7.3

DESIGN OF POST-TENSIONED BEAM FOR SHEAR

Design the shear reinforcement for the beam given in Example 7.1. For the stirrups use Grade R for which $f_{sy.f} = 250$ MPa. The tendon consists of 22 strands of 12.7 mm diameter and from Table 2.1 f_{pb} =1870 MPa and E_p = 195 GPa. The cross-section is shown in Figure 7.8.



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